

### 5.3.14 I/O port characteristics

#### General input/output characteristics

Unless otherwise specified, the parameters given in Table 47 are derived from tests performed under the conditions summarized in Table 17. All I/Os are designed as CMOS and TTL-compliant.

**Note:** For information on GPIO configuration, refer to the application note STM32 GPIO configuration for hardware settings and low-power consumption (AN4899).

**Table 47. I/O static characteristics**

The I/O structure options listed in this table can be a concatenation of options including the option explicitly listed. For instance, TT\_a refers to any TT I/O with \_a option. TT\_xx refers to any TT I/O and FT\_xx refers to any FT I/O. All I/Os are CMOS- and TTL-compliant (no software configuration required). Their characteristics cover more than the strict CMOS-technology or TTL parameters. The coverage of these requirements is shown in the figure below.

| Symbol                           | Parameter                                         | Condition                                                                                       | Min                                       | Typ | Max                                      | Unit |
|----------------------------------|---------------------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------|-----|------------------------------------------|------|
| V <sub>IL</sub>                  | I/O input low level voltage                       | 2.7 V < V <sub>DDIOx</sub> < 3.6 V                                                              | -                                         | -   | 0.3V <sub>DD</sub> <sup>(1)</sup>        | V    |
|                                  | I/O input low level voltage                       |                                                                                                 | -                                         | -   | 0.4 V <sub>DD</sub> - 0.1 <sup>(2)</sup> |      |
| V <sub>IH</sub>                  | I/O input high level voltage                      | 2.7 V < V <sub>DDIOx</sub> < 3.6 V                                                              | 0.7V <sub>DD</sub> <sup>(1)</sup>         | -   | -                                        | V    |
|                                  | I/O input high level voltage                      |                                                                                                 | 0.52V <sub>DD</sub> + 0.18 <sup>(2)</sup> | -   | -                                        |      |
| I <sub>leak</sub> <sup>(3)</sup> | FT_xx Input leakage current <sup>(2)</sup>        | 0 < V <sub>IN</sub> ≤ Max(V <sub>DDXXX</sub> ) <sup>(4)</sup>                                   | -                                         | -   | ±200                                     | nA   |
|                                  |                                                   | Max(V <sub>DDXXX</sub> ) < V <sub>IN</sub> ≤ Max(V <sub>DDXXX</sub> + 1 V) <sup>(5)(6)(4)</sup> | -                                         | -   | ±2500                                    |      |
|                                  |                                                   | Max(V <sub>DDXXX</sub> + 1) < V <sub>IN</sub> ≤ 5.5 V <sup>(5)(6)(4)</sup>                      | -                                         | -   | 750                                      |      |
|                                  | TT_xx Input leakage current                       | 0 < V <sub>IN</sub> ≤ Max(V <sub>DDXXX</sub> + 1 V) <sup>(4)</sup>                              | -                                         | -   | ±200                                     |      |
| R <sub>PU</sub>                  | Weak pull-up equivalent resistor <sup>(7)</sup>   | V <sub>IN</sub> = V <sub>SS</sub>                                                               | 30                                        | 40  | 50                                       | kΩ   |
| R <sub>PD</sub>                  | Weak pull-down equivalent resistor <sup>(7)</sup> | V <sub>IN</sub> = V <sub>DD</sub>                                                               | 30                                        | 40  | 50                                       |      |
| C <sub>IO</sub>                  | I/O pin capacitance                               | -                                                                                               | -                                         | 5   | -                                        | pF   |

- Compliant with CMOS requirements.
- Specified by design - Not tested in production.
- This parameter represents the pad leakage of the I/O itself. The total product pad leakage is provided by the following formula:  
 $I_{Total\_leak\_max} = 10 \mu A + [number\ of\ I/Os\ where\ V_{IN}\ is\ applied\ on\ the\ pad] \times I_{lkg\ max}$
- Max(V<sub>DDXXX</sub>) is the maximum value of all the I/O supplies.
- To sustain a voltage higher than the minimum of V<sub>DD</sub> or V<sub>DDA</sub> plus 0.3 V, disable the internal pull-up and pull-down resistors.
- V<sub>IN</sub> must be less than Max(V<sub>DDXXX</sub>) + 3.6 V.
- The pull-up and pull-down resistors are designed with a true resistance in series with a switchable PMOS/NMOS. This PMOS/NMOS contribution to the series resistance is minimal (~10% order).

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